Beta Regression

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Introduction

Beta regression models continuous outcomes constrained to the open unit interval $(0, 1)$ — proportions, percentages expressed as fractions, biomarker ratios that range from 0 to 1. Ordinary least squares fails on such data: it can predict values outside $(0, 1)$, the residuals are typically heteroscedastic and asymmetric, and confidence intervals on the boundary are unreliable. Beta regression respects the bound, models heteroscedasticity through a precision parameter, and returns interpretable coefficients on the logit scale (or any other supported link).

Prerequisites

A working understanding of the Beta distribution, generalised linear models, and the logit link as a transformation of probabilities to the real line.

Theory

The model parameterises Y as Beta with mean μ and precision ϕ:

$$Y \mid x \sim \text{Beta}(\mu\phi, (1-\mu)\phi),$$

with $\text{logit}(\mu) = x^\top\beta$ and $\phi > 0$ a precision parameter (larger ϕ means tighter concentration around μ). Variable-dispersion beta regression allows ϕ to depend on covariates as well, capturing heteroscedasticity directly. Coefficients are interpretable as log-odds ratios on the proportion: a unit increase in x_j multiplies the odds of "success-vs-failure" of the proportion by $\exp(\beta_j)$.

Values exactly 0 or 1 are excluded from the standard model; common remedies include a small linear shift $(y^* = (y(n-1) + 0.5)/n)$ or a zero-one-inflated beta model that adds explicit point-mass components.

Assumptions

Outcome strictly in $(0, 1)$ (no exact 0 or 1 unless using zero-one-inflated extensions); logit (or chosen) link appropriate; independent observations.

R Implementation

```r
library(betareg)

set.seed(2026)
x <- rnorm(200)
eta <- 0.2 + 0.8 * x
mu  <- plogis(eta)
phi <- 10
y <- rbeta(200, mu * phi, (1 - mu) * phi)

fit <- betareg(y ~ x)
```

```r
summary(fit)

# Predictions on the probability scale
head(predict(fit, type = "response"))
```

Output & Results

`summary(fit)` reports coefficients on the logit scale (with Wald z-tests), the precision parameter $\hat{\phi}$, and pseudo-R^2. `predict(..., type = "response")` returns predictions back on the $(0, 1)$ scale.

Interpretation

A reporting sentence: "Beta regression of the bounded outcome on x estimated $\hat{\beta} = 0.79$ on the logit scale (odds ratio 2.20, 95 % CI 1.79 to 2.71, $p < 0.001$); the precision parameter was $\hat{\phi} = 9.6$, indicating moderate concentration around the conditional mean. A 1-SD increase in x is associated with a roughly two-fold increase in the odds of a higher proportion." Always report both the coefficient and the precision; the precision controls the implied uncertainty.

Practical Tips

- Outcomes exactly equal to 0 or 1 need either a small shift transformation ($y^* = (y(n-1) + 0.5)/n$) or a zero-one-inflated beta model (`gamlss::gamlss(family = BEINF)` or `zoib::zoib`).
- Variable-dispersion beta regression — modelling ϕ on covariates — handles heteroscedasticity directly via `betareg(y ~ x | z)` syntax.
- For longitudinal or clustered proportion data, `glmmTMB(family = beta_family())` adds random effects to a beta GLM.
- The default link is logit; alternatives (probit, cloglog) are available via the `link` argument and rarely change conclusions materially.
- Pearson and quantile residuals are the right diagnostic tools; standard residuals can be misleading for bounded outcomes.
- Compare beta regression against a quasi-binomial GLM with a logit link; the latter is sometimes simpler and gives similar conclusions for moderate dispersion.

R Packages Used

`betareg` for `betareg()` with fixed and variable dispersion; `gamlss` and `gamlss.dist` for zero-one-inflated beta and other bounded distributions; `glmmTMB` for beta GLMMs; `zoib` for fully Bayesian zero-one-inflated beta regression; `DHARMa` for residual diagnostics across non-Gaussian families.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF

- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI